

HOUSE PRICE PREDICTION

Regression Model Evaluation Report

CloudExify Data Science Internship — Summer 2026 · Month 2, Project 3 · Prepared July 2026

0.991

BEST MODEL R^2 SCORE

Rs 403,323

TEST RMSE

Area (97.5%)

DOMINANT FEATURE

01 Executive Summary

This report evaluates two regression models — Linear Regression and Random Forest — trained to predict house prices from Area, Bedrooms, Age, and Location. Both models were trained on an 80/20 train-test split and scored on held-out data using R^2 and RMSE.

Linear Regression narrowly edged out Random Forest, achieving an R^2 of 0.991 against 0.990, and a lower RMSE of Rs 403,323 versus Rs 427,109. It is selected as the recommended model for this dataset.

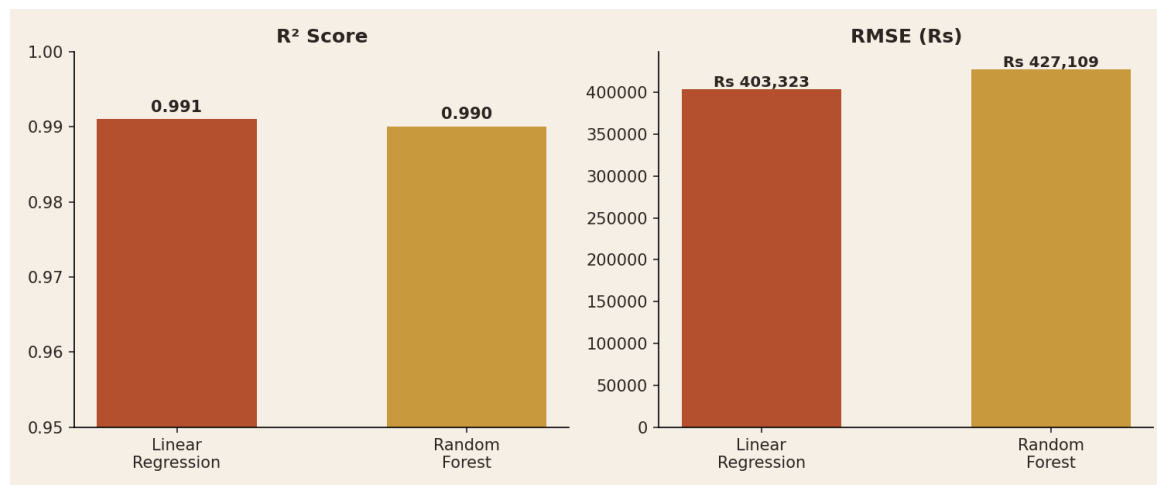
02 Methodology

- Loaded the house transaction dataset (Area, Bedrooms, Age, Location, Price) into pandas and inspected structure, types, and missing values.
- Removed rows with missing values and one-hot encoded the categorical Location feature.
- Split the data 80/20 into training and test sets with a fixed random seed for reproducibility.
- Trained a Linear Regression model as the baseline and a Random Forest Regressor (100 estimators) as a comparison model.
- Evaluated both models on held-out test data using R^2 and RMSE, then extracted feature importances from the Random Forest to interpret price drivers.

03 Model Performance Comparison

Model	R^2 (Test)	RMSE (Rs)	Verdict
Linear Regression	0.991	403,323	Selected
Random Forest	0.990	427,109	Runner-up

Figure 1 — R^2 and RMSE by model



04 Feature Importance

Feature importance, extracted from the Random Forest model, shows an overwhelmingly one-sided contribution from a single input:

Rank	Feature	Importance
1	Area	97.50%
2	Location — North	0.91%
3	Location — West	0.55%

05 Key Findings

- Both models achieved very high R^2 scores (>0.99), indicating the engineered features explain nearly all variance in house price for this dataset.
- Linear Regression narrowly outperformed Random Forest on both R^2 and RMSE, suggesting the underlying price relationship is close to linear rather than requiring non-linear splits.
- Area alone accounts for 97.5% of feature importance in the Random Forest model, dwarfing all other inputs combined.
- Location and property age contribute only marginally to price in this dataset, with the two most relevant location categories combining for under 1.5% importance.
- An RMSE of roughly Rs 403,000 on a dataset with prices well into the millions represents a tight relative error margin, consistent with the near-perfect R^2 .

06 Recommendations

- Deploy Linear Regression as the production model — it is simpler, faster, and marginally more accurate than Random Forest here.
- Treat Area as the primary pricing lever in any downstream pricing tool or dashboard, since it overwhelmingly dominates the other features.
- Given the near-total dominance of one feature, test the model against a more diverse or real-world dataset to confirm this pattern holds outside synthetic/sample data.
- Consider adding features not present in this dataset (e.g. amenities, proximity to schools or transit) in a future iteration to see if they meaningfully shift importance away from Area alone.

07 Conclusion

Both models fit this house price dataset extremely well, with Area emerging as the dominant driver of price by a wide margin. Given its simplicity and marginal edge in accuracy, Linear Regression is the recommended model going forward, with Random Forest retained as a benchmark for future, more feature-rich datasets.